

EmoCare AI — Phase 1 Production PRD

Product Requirements Document: "The Sanctuary" Core Ecosystem Baseline Framework

1. DOCUMENT OVERVIEW & SYSTEM OBJECTIVE

This Product Requirements Document (PRD) establishes the unconstrained technical and design specifications for **Phase 1: The Sanctuary** of EmoCare AI. The core objective of this initial release phase is to cultivate absolute user trust, deep visual obsession, and an unassailable sensory refuge. Every digital layout, macro-component, and data routing system specified in this document enforces our foundational mandate: *"Technology should help people become more human, not less."*

Phase 1 intentionally strips away high-compute analytical noise, focusing entirely on a premium, ultra-low-latency text/voice interactive experience combined with circadian layout fields and private data collection layers.

2. VISUAL ARCHITECTURE & DESIGN TOKENS

To differentiate from standard mobile applications, all rendering targets must eliminate hard card perimeters, solid drop shadows, and high-contrast system modules. The interface floats entirely on an absolute layout stacking structure layer.

THE AMBIENT GLASSMORPHISM SPECIFICATION

- **Background Stacking:** The absolute base layer consists of the GPU-accelerated Skia fragment shader rendering real-time fluid light fields. No visible element background can be 100% opaque white or solid dark tones.
- **The Structural Glass Shell:** Every interactable widget or option button panel container must utilize an explicit blurring fill layer wrapped natively via `<BlurView intensity={25} tint="light">`.
- **The Fine Stroke Accent:** Glass boundaries must render a highly delicate, uniform border stroke tracking `borderWidth: 1.2` and an explicit transparency hex config of `rgba(251, 214, 168, 0.2)`.

- **Typography Restraint:** Core user headings, time tracking blocks, and textual responses use the premium serif family Georgia with letter-spacing padding configurations adjusted to +0.55 pixels.

3. DETAILED FUNCTIONAL REQUIREMENTS & SCREEN SPECS

3.1 THE SANCTUARY HOME WORKSPACE (`HOMESCREEN.TSX`)

The core landing area serves as the somatic anchor of the application workspace layer. It enforces a completely clear, open stacking arrangement where text metrics and operational pills drift organically over the dynamic light shader canvas without solid containers.

- **The Serif Headline Group:** Displays the localized time greeting on line one ("Good evening," or "Good morning,") followed strictly on line two by the individual's name ("Thitsar") with no container boxes.
- **The Subtitle Grounding Line:** A soft italic prose row sitting beneath the user name: *"This space is yours this evening."* styled at a default `opacity: 0.7`.
- **The Central Floating Mascot Capsule:** Houses the minimalist lavender branch graphic enclosed inside concentric pulsing glowing ring fields. Shifting slower than the foreground scroll context during gestures, it creates an intense, 3D parallax spatial layer effect.
- **The Daily Ritual Pill Architecture:** Aligns three distinct capsule nodes horizontally via `flexDirection: 'row'`. All icons inside use uniform Lucide line vector strokes (Sparkles, PenTool, Compass) mapped directly to **Check-In**, **Journal**, and **Breathe** flows. All solid blocks and peach gradient buttons are stripped out in favor of shared glassmorphic transparency tokens.

3.2 LOCALIZED BIOMETRIC CHECK-IN FLOW (`CHECKINSCREEN.TSX`)

An interface constructed to collect explicit personal mood metrics natively without introducing clinical analysis anxiety.

- **The Interactive Mood Grid:** Displays non-judgmental textual representation tags floating natively on the canvas background. Emojis are fully omitted.
- **The Circadian Interpolation Event:** Tapping an option instantly calculates new normalized color node matrix values, passing target parameters to the `SanctuaryAmbientContext`. The Skia canvas smoothly blends its background shaders on the native UI thread to visually map the selected emotional frequency.

3.3 REAL-TIME STREAMING VOICE SANCTUARY (`VOICETALKSCREEN.TSX`)

A full-screen, focused modal overlay that clears out all application tab items and standard navigational header strips, establishing an exclusive audio-first sanctuary environment.

- **The Audio-Reactive Mascot Sphere:** Expands the baseline lavender branch glyph center point into a massive, full-screen fluid Skia blob spanning 70% of the device screen container. The sphere edges organically warp, pulse, and distort using custom vector math equations processing real-time decibel metering inputs from the user microphone stream.
- **The Text Entry Fallback:** Includes a minimal, absolute glass entry bar at the base of the interface container. This ensures full functional operation under Expo Go testing environments by allowing a "Text-In, Audio-Out" workflow during SFSpeech entitlements bypass.

4. TECHNICAL INFRASTRUCTURE & DATA BLUEPRINT

To fulfill **Principle 3: Privacy is Dignity**, data layers must remain completely model-agnostic, ensuring personal context cannot be extracted or processed as transactional training tokens by third-party model suppliers.

SQLITE LOCAL LEDGER ARCHITECTURE

The local mobile client initializes a stateful, encrypted SQLite database instantly upon system generation. Three essential schema matrices dictate the Phase 1 memory data storage lifecycle:

```

-- 1. Explicit Mood Check-In Historical Ledger
CREATE TABLE mood_checkin_ledger (
    entry_id INTEGER PRIMARY KEY AUTOINCREMENT,
    timestamp TEXT NOT NULL DEFAULT (strftime('%Y-%m-%dT%H:%M:%SZ', 'now')),
    mood_label TEXT NOT NULL,
    ambient_progress_target REAL NOT NULL,
    context_excerpt TEXT
);

-- 2. Localized Glassmorphic Journal Document Matrix
CREATE TABLE journal_document_ledger (
    document_id INTEGER PRIMARY KEY AUTOINCREMENT,
    timestamp TEXT NOT NULL DEFAULT (strftime('%Y-%m-%dT%H:%M:%SZ', 'now')),
    journal_prose TEXT NOT NULL,
    associated_mood_id INTEGER,
    FOREIGN KEY(associated_mood_id) REFERENCES mood_checkin_ledger(entry_id)
);

-- 3. Asynchronous Task Energy Matrix Allocation
CREATE TABLE today_triage_matrix (
    task_id INTEGER PRIMARY KEY AUTOINCREMENT,
    timestamp TEXT NOT NULL DEFAULT (strftime('%Y-%m-%dT%H:%M:%SZ', 'now')),
    task_title TEXT NOT NULL,
    energy_category TEXT CHECK(energy_category IN ('Deep Creative Focus', 'Low
Energy Admin', 'Restorative Reflection')),
    completion_status INTEGER DEFAULT 0
);

```

LOW-LATENCY VOICE SOCKET PROTOCOL

The voice room engine relies on a persistent WebSocket socket abstraction interface handling two distinct streaming channels:

- **Upstream Recording Pipeline:** Collects 16,000Hz mono PCM16 audio vectors from the device microphone input. Compresses segments into base64 raw data frames, sending frames over the socket every 100ms. Under Expo Go targets, this channel routes text payloads cleanly through an asynchronous context pipeline.
- **Downstream Synthesis Pipeline:** Processes structural responses coming from the core system intelligence model. Streams incoming audio fragments instantly to the native speaker output stack, using native monitoring thread events to check for immediate barge-in user interruptions. If an user audio queue is registered while Emo is reading a response, the system triggers a localized reset frame to clear the playback cache.

5. PHASE 1 ENGINEERING TRACEABILITY MATRIX

ID TOKEN	COMPONENT CONTEXT	FUNCTIONAL SPECIFICATION RULE	TRACEABILITY FILE
REQ-001	Ambient Canvas Frame	Shader loop must maintain a locked 60 FPS update threshold via GPU hardware.	components/ LivingCanvas.tsx
REQ-002	Interface Capsule Cards	All button items or layout backgrounds utilize light blur styling with champagne gold fine borders.	components/ TodayTriageDashboard.tsx
REQ-003	Daily Activity Navigation	System emojis are entirely banned; navigation buttons route actions through Lucide outline icons.	components/ DailyRitualBanner.tsx
REQ-004	Conversational Triage	Classifier must split text/voice input routing context between Oracle and Sanctuary logic maps.	utils/emoIntent.js
REQ-005	Cognitive Task Framework	Agenda item views organize todos strictly inside custom Energy Groups rather than standard times.	utils/todayTriage.js